Research statement

Kai Liu

1. Introduction

My research lies at the intersection of applied harmonic and functional analysis, quantum information, and quantum computing. I study how mathematical structure governs the representation, reconstruction, and preservation of information. Drawing on frame theory, operator theory, and group representation theory, I connect foundational questions in analysis with the physical behavior of quantum systems, the development of reliable quantum technologies and relevant problems in data science

My research program develops along three connected directions. The first concerns mathematical foundations, particularly how geometry, redundancy, and symmetry determine the capabilities and limitations of signal representations, quantum measurements, and quantum channels. The second connects these foundations to quantum physics through quantum tomography, open-system dynamics, and quantum property preservation. I investigate how evolution, noise, and measurement constraints shape access to quantum information. This perspective also motivates connections to quantum control, including how dynamics and coherent interference can be used to enhance state identification and preserve selected properties.

The third direction develops applications to quantum computing, with extensions to quantum machine learning and data science. Building on the mathematical and physical perspectives above, I aim to investigate how measurement design, reconstruction, and information preservation can support computation, estimation, and learning under noise and limited measurements. This direction connects my theoretical work with broader questions about quantum advantages, the efficiency and reliability of quantum methods and their potential roles in data analysis.

2. Summary of Previous Research Contributions

In previous work, I have developed structural criteria for quantum-state identification and information preservation, together with frame constructions for robust reconstruction. The following four themes connect these contributions to quantum tomography, communication and error correction, noisy quantum dynamics, and signal analysis.

2.1 Quantum Tomography and the Geometry of Measurements

Quantum tomography depends on whether the available measurements determine an unknown state. For pure states, the physically irrelevant global phase leads naturally to phase retrieval, a central inverse problem in applied harmonic analysis. In [4], we developed a framework for determining when measurements of a quantum channel's output retain enough information to identify its pure input. The adjoint channel connects output observables to measurements on the input, allowing frame theory and operator methods to address this question.

We obtained an exact characterization for channels with two Kraus operators and developed criteria for higher Kraus rank through relative spectra of operator tuples. We also constructed phase retrievable channels with prescribed Choi rank and, under specified assumptions, minimal numbers of rank-one observables. These results show how pure-state identification can remain possible through channels that lose other information, providing a mathematical foundation for tomography that exploits prior knowledge of the input state.

The underlying measurement geometry is also the subject of my work on frame dimension functions [3].

We related these functions to the redundancy of phase retrieval measurements, including frames that do not identify every vector. In particular, we characterized exact phase retrieval redundancy over real Hilbert spaces through dimension function exactness and determined the dimension ranges of full spark frames. This work gives a more detailed account of measurement structure than the existence of unique reconstruction alone. Our work [8] synthesizes the connections between frames and quantum channels and clarifies the relationships among pure-state identification, full tomography, perfect discrimination, and exact correction.

2.2 Symmetry, Quantum Communication, and Error Correction

Symmetry provides mathematical structure for determining which information survives a quantum channel. In [1], we studied identification modulo a group action: an invariant channel identifies pure-state orbits when equal outputs occur precisely for states related by the symmetry. For finite groups acting on complex Hilbert spaces, we proved a rigidity result: such identification is possible only when the group acts trivially on pure states, in which case orbit identification reduces to ordinary pure-state identification. We also obtained representation-theoretic criteria and constructions in compact group settings that permit nontrivial orbit identification. These results reveal how the choice of symmetry constrains the information an invariant channel can retain.

In [2], we connected representation structure directly to quantum communication and error correction for twirling channels, which average over a group action. If the inequivalent irreducible components have multiplicities $m_1, \ldots, m_d$, we showed that the maximum number of classical messages distinguishable without error in one channel use is $\sum_i m_i$, while the largest exactly correctable quantum code dimension is $\max_i m_i$. We also determined the zero-error classical capacity and obtained bounds for subspaces on which pure states remain identifiable. These formulas make symmetry data an explicit tool for analyzing communication and encoded quantum information, connecting representation theory and operator algebras with quantum coding theory.

2.3 Quantum Property Preservation and Coherent Interference

A further part of my work examines which relationships between quantum states survive noise. In [5], we studied exact preservation of fidelity, a measure of similarity between states, under mixed unitary channels. We derived criteria for preserving perfectly distinguishable pure state pairs and obtained a necessary and sufficient condition for nonorthogonal pairs under a two-unitary mixture. For nonunitary qubit phase damping channels, we characterized families whose pairwise fidelities remain unchanged even though their individual states undergo decoherence. These results identify preserved relationships through the geometry of the states and the relative unitary operators describing the noise. For nonzero fidelity, they also provide explicit equality cases for the data processing inequality of the sandwiched Rényi divergence of order one-half.

In [6], we investigated how coherent interference can enhance state identification. Using complementary channels and operator-valued frames, we developed a structural characterization of phase retrievability and studied coherent recombination of two channel arms. For specified interferometric implementations, we constructed examples in which neither arm identifies all pure input states, but the resulting port map does. The interference cross terms introduce additional information, and quantitative injectivity indices describe the resulting state separation. This work connects operator and frame theory with the coherent control of quantum processes, showing how a physical mechanism can improve the identification properties of a measurement arrangement.

2.4 Structured Frames and Robust Signal Reconstruction

My work also develops frame theory as a foundation for reconstructing signals from incomplete measurements. A redundant representation provides flexibility to impose algebraic constraints and choose a reconstruction system adapted to coefficient loss. In [7], we developed a framework connecting prescribed vanishing moments, dual frames, and dilations. A central result characterizes the existence of a dual satisfying these moment constraints through the relative position of the constraint space and the range of the frame's analysis operator.

We proved that every frame of $N = n + k$ vectors in dimension n admits a dual robust to any k coefficient erasures. Using a suitable dual for encoding, we recover ℓ missing coefficients through moment relations by solving a system of size ℓ under appropriate nonsingularity conditions. We also developed approximate recovery and iterative refinement under restricted isometry assumptions. Our framework further characterizes balanced frames equivalent to graph frames: each linearly connected component must contain exactly one more vector than the dimension of its span. These results connect harmonic analysis and spectral graph theory with robust reconstruction from incomplete data.

3. Current Research Projects

My current research extends my work on frames and quantum channels through three ongoing projects concerning dynamical tomography, the resources required to realize quantum processes, and the distribution of information across quantum systems. Each project connects structural questions in mathematical analysis with constraints arising from measurement, physical evolution, or information protection.

3.1 Dynamical Quantum Tomography and Measurement

I am investigating how a fixed measurement scheme can identify a pure state from repeated preparations observed at different times under a known quantum evolution. The central question concerns the tradeoff between the number of resolved measurement outcomes and the number of observation times. I study when additional temporal observations can compensate for restricted readout, and when symmetry or features of the dynamics prevent this.

Building on my work on channel phase retrievability [4], I combine frame theoretic ideas with observability methods for open quantum systems, including Lindblad dynamics. I am particularly interested in how coherent evolution and dissipation affect accessible information, and how sampling choices influence the stability of reconstruction. This project connects applied harmonic analysis with quantum tomography and open system physics, with the goal of understanding measurement resources under concrete physical constraints.

3.2 Noncommutative Graphs and Quantum Channel Complexity

A channel's noncommutative graph describes which inputs can be distinguished without error. I am studying how this structure constrains the resources required to implement a quantum process in successive stages. My focus is post-processing complexity, measured by the smallest intermediate system dimension compatible with prescribed confusability structures before and after post-processing.

A central question concerns a split property: whether the relevant operator system admits a tensor product form compatible with a prescribed subsystem decomposition. I investigate how this property relates to implementation costs, particularly in the presence of protected quantum information. I am also examining which features of a channel affect these requirements beyond its graph, and how the picture changes when approximate implementation is allowed. Building on my work on quantum codes and twirling channels [2], this project connects operator theory and operator algebras with quantum error correction and the analysis of implementation resources.

3.3 Frame Theory and Quantum Masking

I am exploring how frame theory can describe quantum masking for prescribed finite families of pure states. The goal is to encode a state family into two subsystems while preserving its inner product geometry and making each subsystem alone reveal no information about which state was encoded. This raises questions about how information retained in global correlations can be characterized through the geometry of the original state family.

Using Gram matrices and frame operators, I investigate how redundancy and state overlaps constrain possible masking arrangements and their entanglement requirements. I am also studying approximate masking, with an emphasis on how perturbations affect information leakage to an individual subsystem and the information accessible through joint measurements. This project extends my work on frame geometry and constrained representations [3, 7] toward the mathematical foundations of quantum information protection, connecting harmonic analysis with entanglement and state discrimination.

4. Future Research Directions

My long-term goal is to develop mathematical principles that make the representation, acquisition, and protection of information more reliable, with applications to quantum technologies, machine learning, and data science.

4.1 Mathematical Foundations of Quantum Information and Computing

I aim to advance the mathematical foundations of quantum information theory and quantum computing through connections with applied harmonic analysis, operator theory, operator algebras, and group representations. Building on my work on frames and quantum channels, I plan to explore the geometric and algebraic principles underlying quantum states, measurements, and operations. A central theme is understanding how symmetry, redundancy, and noncommutative structure govern the representation, transformation, and preservation of quantum information.

This direction will also develop connections among quantum measurement theory, information protection, and the mathematical theory of quantum computation. I am interested in establishing analytical tools that clarify the capabilities and limitations of quantum processes, the role of quantum resources, and the principles underlying reliable computation. These broader goals will guide the development of mathematical frameworks that connect different quantum information tasks and reveal common structural foundations. Questions arising from quantum information will, in turn, motivate new developments in harmonic analysis and operator theory, allowing this program to contribute both to mathematics and to the theoretical foundations of quantum technologies.

4.2 Mathematical Principles for Quantum Realization

I aim to explore how mathematical structures in quantum information acquire physical meaning and guide the realization of quantum processes. Drawing on frame theory, operator theory, and group representations, I plan to investigate how geometry, symmetry, and algebraic constraints relate to physical mechanisms such as interference, entanglement, and dissipation. My work on fidelity preservation and interferometric coupling [5, 6] provides starting points for developing these connections between mathematical analysis and quantum physics.

I plan to pursue this direction through physical models, open system dynamics, and quantum control. A central interest is how controlled interactions and system environment couplings can realize mathematically specified measurements, channels, and encodings under constraints on available controls and resources. Models of interacting spins and light matter coupling offer settings for examining how mathematical structure can inform physical constructions and the preservation of quantum properties. Physical

phenomena and implementation constraints will also motivate new questions in harmonic analysis and operator theory. Through this exchange, I aim to develop mathematical tools that explain quantum behavior and suggest approaches to the realization and control of quantum systems.

4.3 Quantum Applications, Machine Learning, and Data Science

As quantum computers and related technologies develop, I aim to investigate how quantum representations of data can support reliable learning and inference. A natural starting point is quantum kernel learning, in which similarities between encoded data determine a learning model. Drawing on my work on symmetry and fidelity preservation [1, 5], I plan to investigate how noise changes the geometry of quantum data representations and how those changes affect learning accuracy. I am especially interested in identifying structured encodings for which useful similarities can be estimated reliably with limited measurements.

Frame theory offers a complementary route through the design of estimators for selected observables and features. I aim to connect measurement design and reconstruction error with statistical guarantees for learning from quantum experiments. In parallel, I will pursue classical applications of constrained frame representations to feature estimation and reconstruction from incomplete data. A further connection to artificial intelligence is the use of learning methods to adapt measurement schedules and control choices, guided by the observability and stability principles developed in the first two directions.

A longer term objective is to identify conditions under which quantum resources provide an advantage in learning or inference. I will compare proposed methods with appropriate classical approaches, accounting for data access, state preparation, repeated measurements, noise, and computational cost. The aim is to establish where quantum structure offers measurable benefits and to translate those benefits into reliable methods for computation and data analysis.

References

[1] K. Liu, C. Cheng, and D. Han. "Orbit-injective covariant quantum channels." Linear Algebra and its Applications 668 (2023): 28--50.

[2] D. Han, and K. Liu. "Zero-error correctibility and phase retrievability for twirling channels." Reports on Mathematical Physics 93.1 (2024): 87--102.

[3] D. Han, and K. Liu. "Frame dimension functions and phase retrievability." Advances in Operator Theory 9.2 (2024): 31.

[4] K. Liu, and D. Han. "Phase retrievability of frames and quantum channels." Linear Algebra and its Applications 725 (2025): 378--400.

[5] K. Liu, and D. Han. "Fidelity preservation for mixed unitary quantum channels." arXiv:2410.20117 (2024). Revision submitted, and under review by QIP.

[6] K. Liu, D. Han, and O. Nour. "Enhancing phase retrievability of quantum channels via interferometric coupling." arXiv:2604.24363 (2026). Revision submitted, and under review by Phys. A.

[7] D. Han, S. Mondal, O. Nour, and K. Liu. "Vanishing moment dilations and applications to balanced and graph frames." Manuscript under review by JFAA.

[8] D. Han, and K. Liu. "Frame phase retrievability and state distinguishability of quantum channels." arXiv:2608.30006 (2026). Manuscript under review as a book Chapter.

[9] S. Mondal, D. Han, and K. Liu. "Extreme Points of Optimal Dual Frames for Erasures." (Under review by Journal SaSiDa)